

Comparing Model Architectures with k-Fold Cross-Validation

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A performs better than B?

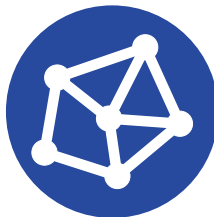
Architecture

A



Architecture

B



VS

Dataset



?

Problem

Goal: determine whether **Architecture A performs better than Architecture B.**

Method: **k-fold cross-validation + t-test**

Objective:

- Obtain multiple performance estimates
- Perform a **statistical comparison between architectures**

k-Fold Cross-Validation

 Dataset is divided into K folds. If $K = 5$, then

fold1 fold2 fold3 fold4 fold5

Iteration 1 **val** **train** **train** **train** **train**  **score 1**

k-Fold Cross-Validation

 Dataset is divided into K folds. If $K = 5$, then

	fold1	fold2	fold3	fold4	fold5	
Iteration 1	val	train	train	train	train	→ score 1
Iteration 2	train	val	train	train	train	score 2
Iteration 3	train	train	val	train	train	score 3
Iteration 4	train	train	train	val	train	score 4
Iteration 5	train	train	train	train	val	score 5

K-Fold Cross-Validation

Example with $K = 5$.

Validation fold	 Score A	 Score B
1	a_1	b_1
2	a_2	b_2
3	a_3	b_3
4	a_4	b_4
5	a_5	b_5

$$\mu_A = \frac{1}{K} \sum_{k=1}^K a_k \quad \wedge \quad \mu_B = \frac{1}{K} \sum_{k=1}^K b_k$$

Comparison

Architecture

A



Architecture

B



$$\mu_A > \mu_B$$

One-sided paired t-test

Requirements: Normality condition of analyzed data

Paired Comparison

Because both models use the **same folds**, the comparison is **paired**. For each fold we compute the difference: $d_k = a_k - b_k$

Validation fold	Score A	Score B	d_k
1	0.82	0.80	0.02
2	0.79	0.76	0.03
3	0.83	0.81	0.02
4	0.81	0.78	0.03
5	0.80	0.78	0.02

Statistical analysis is performed on the **vector of differences**:

$$d = [d_1, d_2, d_3, d_4, d_5]$$

Normality Test (Shapiro-Wilk)

The one-sided paired t-test assumes that the differences follow a normal distribution. We test the normality of the differences vector:

$$d = [d_1, d_2, d_3, d_4, d_5]$$

```
import numpy as np
from scipy import stats

ScoreA = np.array([0.82, 0.79, 0.83, 0.81, 0.80])
ScoreB = np.array([0.80, 0.76, 0.81, 0.78, 0.78])
d = ScoreA - ScoreB

w, p = stats.shapiro(d)
if p > 0.05: # W close to 1.0 indicate normality.
    print("Differences are normally distributed")
```

One sided paired t-test

We test whether the **mean difference** is greater than zero.

Null hypothesis:

$$H_0 : \mu_A \leq \mu_B$$

Alternative hypothesis:

$$H_1 : \mu_A > \mu_B$$

Apply a **one-sided paired t-test** to test whether $\mu_A > \mu_B$.

Decision rule:

$$p < 0.05 \Rightarrow \text{significant difference}$$

One-sided paired t-test in Python

If the differences are normally distributed, we apply a one-sided paired t-test.

Python example:

```
import numpy as np
from scipy import stats

A = np.array([0.82, 0.79, 0.83, 0.81, 0.80])
B = np.array([0.80, 0.76, 0.81, 0.78, 0.78])

t, p = stats.ttest_rel(A, B, alternative="greater")

if p < 0.05:
    print("A performs significantly better than B")
```

Conclusion

Procedure:

- 1 Train models using **k-fold cross-validation**
- 2 Obtain **k scores per architecture**
- 3 Check normality distribution of **fold-wise score differences**
- 4 Apply a **one-sided paired t-test** to get the p value

if $p < 0.05$ **then**

 | Architecture A performs better than Architecture B ;

else

 | No significant difference;
